CALCULUS I 2019 Spring Midterm Exam	Dept. or School		Year		proctor	
	Student ID		Name			

※ Your answer must be provided with descriptions how to get the answer.

1. (a) (2 points) Find the exact value of the following expression.

$$\cosh\left[\tanh^{-1}\left(\frac{1}{2}\right)\right]\times\cosh\left[\tanh^{-1}\left(\frac{1}{3}\right)\right]-\sinh\left[\tanh^{-1}\left(\frac{1}{2}\right)\right]\times\sinh\left[\tanh^{-1}\left(\frac{1}{3}\right)\right].$$

(b) (3 points) If $f(x)=\sqrt[4]{\frac{1+\tanh x}{1-\tanh x}}-\sinh\left(\frac{x}{4}\right)\cosh\left(\frac{x}{4}\right)$, find $f'(\ln 9)$.

2. (3 points) (a) Find $\lim_{x\to\infty}(\ln x)^{\frac{\ln x}{x}}$.

(b) (2 points) Find linearization $L(x)$ of $f(x)=\sec x$ at $a=0$ and use it to estimate $\sec\left(\frac{1}{\sqrt{2019}}\right)$.

3. **(4 points)** Find $\lim_{x \to 0^+} \frac{1}{\sqrt{x^3}} \int_{\sqrt{x}}^{2\sqrt{x}} \sin(t^2)\, dt$.

4. **(4 points)** The solution between 3 and 4 of the equation $\tan x = 0$ is π. Starting with $x_1 = 3$, find the second approximation x_2 to π. (Use $\sin 6 = -0.30$)

5. Let $I_n = \int_0^1 (1-x^2)^n dx$ where n is a positive integer.

(a) **(2 points)** Find the number k such that $I_{20} = k I_{19}$.

(b) **(3 points)** Find I_{20}.

6. (4 points) Suppose the graph of the function $y = x^3 + px^2 + qx$ intersects the parabola $y = x^2$ when $x = 0$, $x = 1$ and $x = a$, where $a > 1$. If the two regions between the curves have the same area, find the volume of the solid obtained by rotating the two regions about the y-axis.

7. (3 points) Evaluate the indefinite integral:

$$\int \frac{1}{x\sqrt{3x^2 - 2x - 1}}\,dx.$$ (**Hint**: Substitute $x = \frac{1}{t}$)

8. (5 points) Evaluate the definite integral:

$$\int_0^{\pi}\left[\frac{\cosh x}{\cosh x+\cosh(\pi-x)}+\frac{x\sin x}{1+\cos^2 x}\right]dx.$$

9. (5 points) Evaluate the indefinite integral:

$$\int\left(\frac{1}{\sin^2 x\cos^2 x}+\frac{\sec^2 x}{\tan^3 x-1}\right)dx.$$